

Project Design Phase
Problem – Solution Fit

Date	18 th June 2026
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Project Name	ShopEZ – E-Commerce Application
Maximum Marks	2 Marks

Problem – Solution Fit Template:

1. Target Customer

User 1 – Online Shopper

Regular individuals who want to browse and purchase products online in a simple, fast, and secure manner without going through complicated processes.

User 2 – Admin / Store Manager

Business owners or sellers who need to manage their product listings, track customer orders, handle categories, and monitor business performance from a single platform.

2. Problems

For Online Shoppers:

- Difficulty finding the right products due to unorganized and cluttered product catalogs
- Long and confusing checkout process that leads to cart abandonment
- No instant order confirmation after placing an order
- Lack of trust in platform security while entering payment and personal details
- Unable to track order history or current order status easily
- Missing or incomplete product descriptions, images, and customer reviews

For Admin / Seller:

- No centralized dashboard to manage products, orders, and categories together
- Manual and time-consuming process of updating product listings
- No analytics or reporting tool to track sales and business performance
- Difficulty managing banner images and product categories efficiently
- No role-based access control to separate admin and user permissions
- Errors and delays caused by lack of a proper order management system

3. Existing Alternatives

- Large platforms like Amazon and Flipkart are too complex and overwhelming for small businesses

- Basic shopping websites lack proper admin dashboards and analytics features
- Manual tools like spreadsheets and notebooks are error-prone and not scalable
- Generic e-commerce templates do not offer role-based access or custom analytics
- Existing platforms charge high fees and do not provide personalized management features

4. Solution

ShopEZ is a full-stack MERN e-commerce web application that solves both user and admin problems in one platform:

For Online Shoppers:

- Clean and organized product catalog with detailed descriptions, images, pricing, and customer reviews
- Simple "Shop Now" button on every product for quick and easy order initiation
- Secure and fast checkout process with shipping address and payment details entry
- Instant order confirmation with complete order summary after successful purchase
- Dedicated profile section to view order history and track current orders
- JWT-based secure authentication to protect personal and payment information

For Admin / Seller:

- Centralized admin dashboard with complete control over products, orders, and categories
- Easy product management — add, update, and delete product listings in a few clicks
- Banner image and category management to keep the store organized and visually appealing
- Recharts-powered analytics to monitor sales trends and business performance
- Role-based access control to ensure only authorized admins can access sensitive features
- Real-time order tracking and management to reduce errors and improve customer satisfaction

5. Key Differentiators

- Built using the MERN Stack (MongoDB, Express.js, React.js, Node.js) for fast and scalable performance
- Separate authentication for users and admins with JWT token-based security
- Recharts integration for visual and interactive sales analytics in the admin dashboard
- Simple and intuitive interface designed for both shoppers and non-technical store managers
- Role-based access control ensuring data security and proper permission management

- One unified platform combining shopping experience and store management tools

6. Unique Value Proposition

ShopEZ is a one-stop full-stack e-commerce platform that gives online shoppers a smooth, secure, and fast shopping experience while giving admins and sellers a powerful, easy-to-use dashboard to manage their entire store — all in one place, built with modern MERN stack technology.

7. Metrics to Measure Success

Metric	Goal
User Registration & Login	Increase number of registered users
Cart to Order Conversion	Reduce cart abandonment rate
Order Confirmation Rate	100% order confirmation after checkout
Admin Product Management	Reduce time taken to add or update products
Sales Analytics Usage	Admins checking dashboard regularly for decisions
Customer Satisfaction	Positive feedback on shopping experience